

Definition and first terms

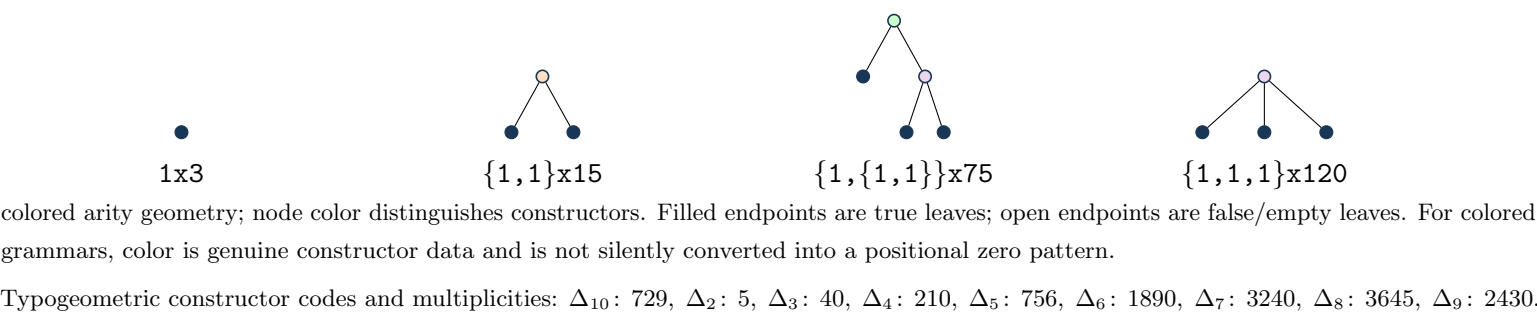
$37 A(x) = 36 + 81 x + A(x)^{10}$, with the origin-normalized solution.

Normalization/grammar: $A(x)=1+(3)*T(x)$; T-polynomial coefficients are the Delta multiplicities below.

$a(0), \dots, a(7) = 1, 3, 15, 270, 5505, 124818, 3028200, 76896180$.

Kernel used throughout: $\rho(u) = u(1 - 5 u^1 - 40 u^2 - 210 u^3 - 756 u^4 - 1890 u^5 - 3240 u^6 - 3645 u^7 - 2430 u^8 - 729 u^9)$, so $x = \rho(T(x))$.

Typogeometry



Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \dots, k_9) \in \mathbb{Z}_{\geq 0}^9 : \sum_{j=1}^9 j k_j = n - 1\}, \quad K = \sum_{j=1}^9 k_j,$$
$$a(n) = \frac{3}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_9} 5^{k_1} 40^{k_2} 210^{k_3} 756^{k_4} 1890^{k_5} 3240^{k_6} 3645^{k_7} 2430^{k_8} 729^{k_9}.$$
$$a(n) = \frac{3}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^9 P_r(n) a(n+r) = 0$ for $n \geq 1$. The exact degree-9 coefficient polynomials are embedded in `certificate_payload.json` (source: `data/recurrence.json`).

Differential equation

The scalar linear ODE has order 9. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `data/ode.json`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n, u) = N(n, u) / \rho(u)^8$, $\deg_u N = 81$.

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.

Matrix dimensions: 20×20 ; remainder 9×10 , rank 9, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.